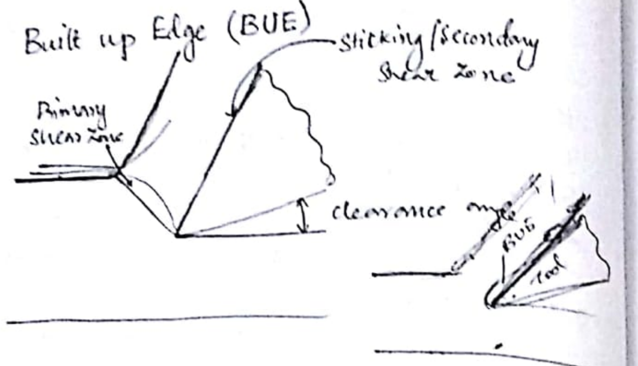
Built up Edge (BUE)



When the zone under cutting action is examined the following observations can be made. The uncut layer deforms into a chip after it goes through a severe plastic deformation in the primary shear zone. Just after its formation the chip flows over the rake surface of the tool and the strong adhesion b/w the tool and newly formed chip surface results in some sticking. Thus the chip material at the surface undergoes a further plastic deformation since despite the sticking it flows. This zone is reffered to as the secondary shear zone.

Under suitable conditions the maching operation is smooth and stable and produces continuous ribbon like chips.

At somewhat high speed the temperature increases and the tendency of the plastically deformed material to adhere to the rake surface increases and a lump is formed at the cutting edge. This is called a builtup edge. It grows up a certain size but ultimately breaks due to the increased force excerted on it by the adjacent flowing material. After it breaks broken fragments adhere in the finish surface and chip surface results in a rough finish.

Low speed, depth of cut is more, brittle material

(i) Continuous chip without built up edge

Conditions are: ductile material
Small uncut thickness
high cutting speed
Large rake angle
Suitable cutting feed

(ii) Continuous chip with built up edge

Stronger adhesion b/w chip and tool face
Low rake angle
Large uncut thickness

Discon

(iii) Discontinuous Chip

Low cutting speed
Brittle work material
Small rake angle
Large uncut thickness

Materials commonly used for making cutting tools are

High carbon steel
High speed steel
Cemented carbide
Ceramics

For grinding and other machining processes abrasive materials like silicon carbide, aluminium oxide, diamond are used.

Composition ~~of high speed steel~~

Material	C	W	Cr	V	Mn	Fe
Carbon tool steel	0.9				0.6	Rest 98.5
High speed steel	0.75	18	4	1	0.6	Rest 75.65

Cemented carbide is produced by powder metallurgy technique. For this powders of several carbide compounds are pressed and bonded together in a matrix to form the cemented material. The matrix is always cobalt.

WC (Tungsten carbide)
Co
TaC (Tantalum carbide)
TiC (Titanium carbide)

Tool Life and Machinability

Major criteria for judging machinability are:

(i) Machining ~~process~~ forces and power consumption

A machining requiring a large force indicates low machinability and vice versa

(ii) Surface finish

(iii) Tool life

The length of the period for which a tool can be used is defined as tool life. This criterion is also linked up to the productivity and economics and can be a very good index for an overall judgement of a machining operation.

As long as the available power is enough and doesnot cause any problem on the machining and we are not concerned with a very special precision machining operation requiring a high quality of finish, the tool life may be considered the index for judging the machining ability. In cases ~~where~~ the major objective is to remove quickly and cheaply ~~the tool~~ life is used as a direct measure of machinability.

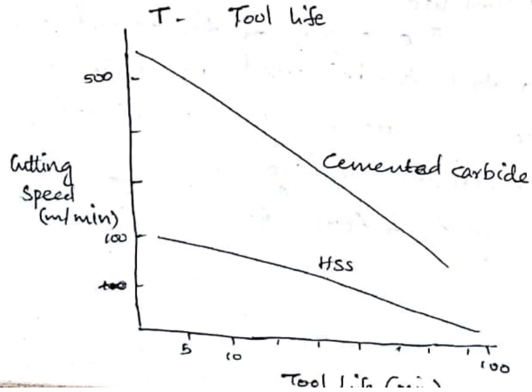
Tool life equation

$$VT^n = C$$

where n, C - constants depending on tool and work material

V - cutting speed

T - Tool life



Though the cutting speed is the most dominant variable the other cutting parameters, i.e, the uncut thickness and width of cut also affects tool life. When the machining is through the production of continuous chips without built up edge, the generalised Taylor equation is,

$$V = \frac{C'}{T^n t^p w^q}$$

where C, n, p, q - constants

t - uncut thickness

w - width of cut

T - Tool life

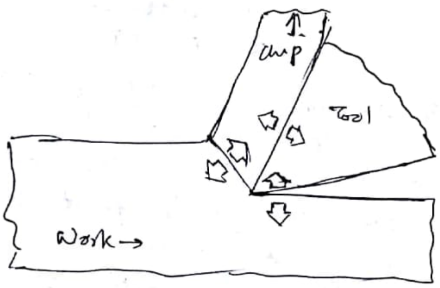
Heat Generation and Cutting Tool Temperature

When a material is deformed ~~elastic~~ elastically some energy is spent to increase its strain energy ~~to~~ which it is returned after unloading. But in plastic deformation most of the energy ~~does~~ thus spent is converted into heat.

During machining the plastic deformation is large and takes place at a very high rate and under such condition almost 99% of the energy is converted into heat. For cutting ~~a~~ low strength material heat generation and the consequent increase ~~in~~ temperature of the ~~too~~

and cutting zone is not a big problem. However when ferrons and high strength materials are machined, the temp raises with the speed and the tool strength decreases. leading to a faster wear and failure. So though machining at a high speed is desirable for higher productivity the faster tool wear due to the high temp puts a limit to the cutting speed.

Sources of Heat Generation



The major plastic deformation takes place in the shear zone and this heat primary heat source. A large fraction of this heat goes to the chip. The sliding motion of the chip on the rake surface of the tool also generates heat and this is the secondary heat source. From this source also the chip takes away the major portion of the heat. There is another source of heat where the job rubs the against the flank surface of the tool. But with sharp tools the contribution of this zones to the heating phenomenon is insignificant.

The total power consumption

Total rate of heat generation

$W = F_c V$

$= W_p + W_s$

$W_s = F V_c = F_r V$

$W_s = W - W_p$

$F V_c = F_c V - W_p$

$W_p = F_c V - F V_c$

When a material particle goes across the primary deformation zone the temp is given by $\theta_p = \frac{(1-\Delta) W_p}{\rho c V t_1 w}$ where

Δ is the fraction of primary heat which goes to the workpiece

ρ – Density of material

t_1 – uncut thickness

w – width of cut

c – specific heat of material

Since the computation of Δ ~~detail~~ needs an elaborate analysis you will simply find the result which agrees well ~~theory~~ with theory results. It has been found the Δ is a function of shear angle ϕ and nondimensional quantity

$$\theta = \frac{\rho c v t_1}{k}$$

k is the thermal conductivity

Determine the maximum temp along the rake face of the tool when machining mild steel given

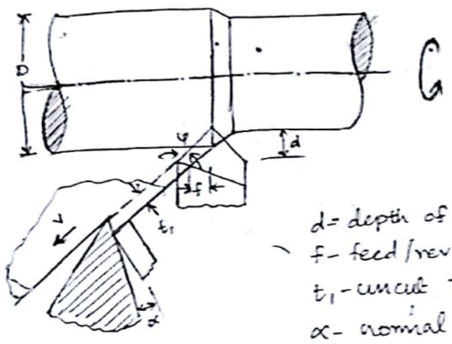
Shear strength of the material 400×10^6 N/m², $\alpha = 0$,

$v = 2$ m/s, $t_1 = 0.25$ mm, $w = 2$ mm, $\mu = 0.5$

$\rho = 7200$ kg/m³, $k = 43.6$ W/m °C

$c = 502$ J/kg °C, $\theta_0 = 40$°C. Using the Lee and Shaffer shear angle relationship

Turning Tool - Single Point



d = depth of cut

f - feed/rev

t_1 - uncut thickness

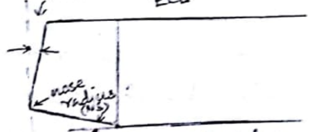
α - normal rake angle

ψ - side cutting edge angle

$t_1 = f \cos\psi$

$w = \frac{d}{\cos\psi}$

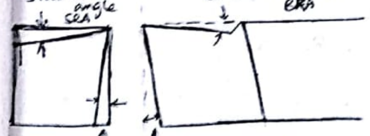
$V = \pi D N$



Total signature

6° - 10° - 7° - 7° - 10° - 30° - 0.5mm

BRA SRA FC SC ECE SCE NS



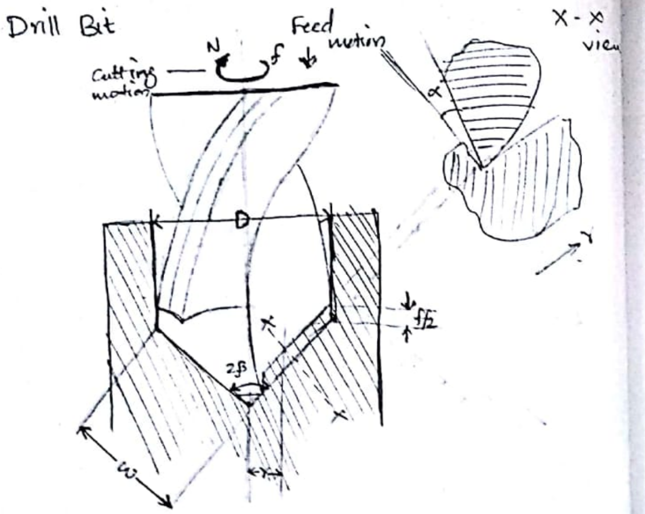
Power, $W = F_c V$

Cutting time, $T_c = \frac{L}{fN}$

Material Removal Rate

MRR = fdv

Drill Bit



The most common hole making operation is drilling and it is usually performed with the help of a twist drill. Unlike shaping and turning this involves two principal cutting edges. If the total advancement of the drill per revolution (the feed rate) is f then the share of each cutting edge is f/2.

Uncut thickness, $t_1 = \frac{f}{2} \sin\beta$

Width of cut, $w = \frac{D}{2} \sin\beta$

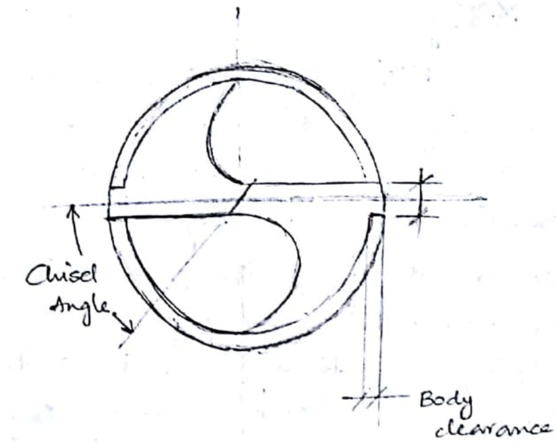
D - Diameter of drill bit

β - Half cone/point angle

$$\alpha \approx \tan^{-1} \frac{(2r/D)\tan\psi}{\sin\beta}$$

r - ~~small~~ nominal radius at the cutting edge

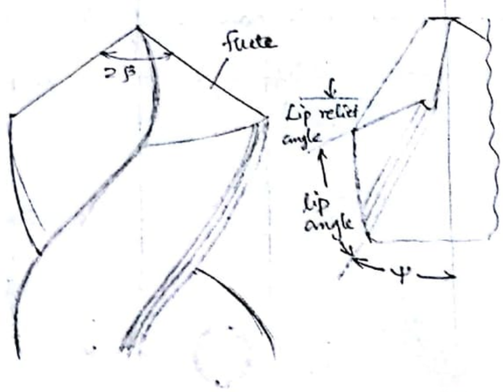
D - nominal diameter of drill bit ψ - Helix angle



Torque,

$M \approx 0.6 F_c D$

$F \approx 5 F_T \sin\beta$



Estimate the torque and thrust force when drilling a solid block of mild steel with a normal twist drill. Given that D = 20mm, N = 240 rpm, τ_s = 400 N/mm², f = 0.25 mm/rev, $2\beta = 118°$, $\psi = 30°$, $r = D/4$

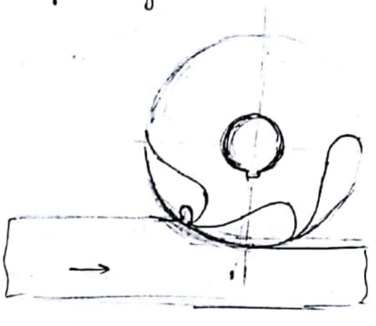
$$\alpha = \tan^{-1}\left[\frac{(2r/D)\tan\psi}{\sin\beta}\right] = \tan^{-1}\left[\frac{(2\times5/20)\tan 30}{\sin 59°}\right]$$

$$= 18.61°$$

Milling

Milling is perhaps the most versatile machining operation and most of the shapes can be generated by this operation. It is especially more indispensible for machining the parts without rotational symmetry. Unlike, turning, shaping and drilling tools the milling tool posses a large number of cutting edges. The shaft on which the cutter is mounded is commonly known as the arbor. Milling operations can be classified into two groups - horizontal and vertical

Upmilling

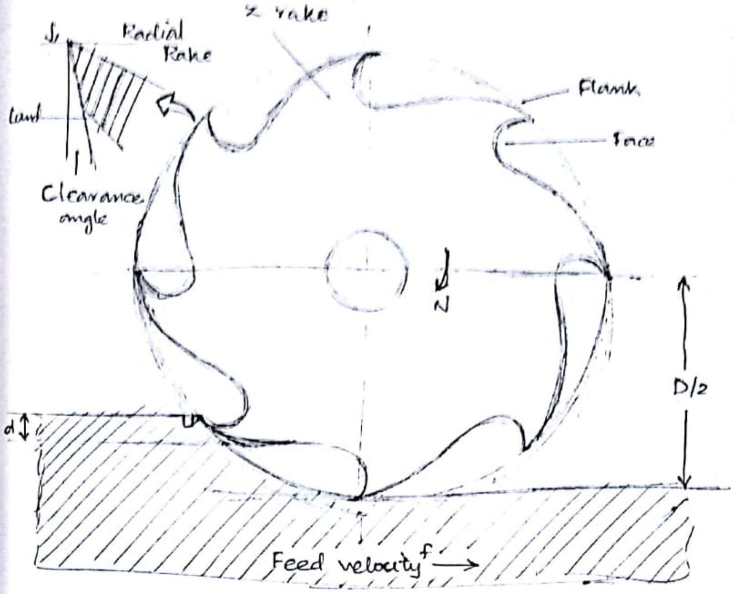


Downmilling

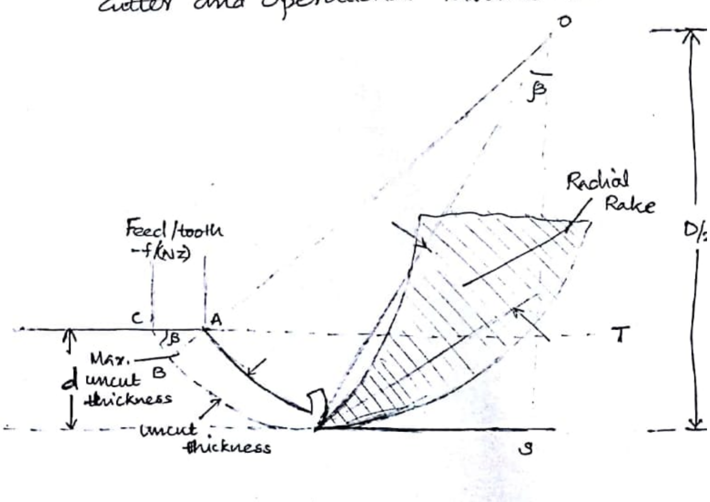
Both in same direction

Milling cutter

Tool Geometry



Cutter and Operational Parameters



Details of chip Formation